

PV INTEGRATED EV SYSTEM

EE 405

Prepared by:

[ATHIYAMAN S. : E/21/043]

[BHASHINI A.J. : E/21/060]

[SHAKTHIDHARAN S. : E/21/376]

Supervised by:

[Dr. PRABATH J. BINDUHEWA]

[Department of Electrical and Electronic Engineering]

[University Of Peradeniya]

Date: 05.08.2026

TABLE OF CONTENTS

Week 01,02

1. Timeline and Gantt chart	3
2. System Integration and Range Anxiety	
2.1 System Integration	4
2.2 Battery Type Selection	5
2.3 Reduction of Range Anxiety	6
3. PV Panel Selection and MPPT Design	
3.1 PV Panel Selection	7
3.2 Maximum Power Point Tracking (MPPT)	8
3.3 Daily Energy Generation Calculation	8
4. Converter from PV to EV Battery	
4.1 Topology Selection	9
4.2 Parametric Calculations	10
5. References	12

TIMELINE

ACTIVITY	DELIVARABLE
1.Literature review on vehicle-integrated PV systems (Week 1-2)	Literature Review Report (week 03)
2.Developing the irradiance meter and artificial light testing setup (week 3-8)	Functional Irradiance and Temperature Meter (week 04)
	Constructed Controlled Testing Box with incandescent bulb array (week 09)
	Sensor Calibration and Design Report (week 09)
3.Simulating the full-scale EV converter and closed-loop MPPT logic (week 3-10)	Comprehensive Simulation Validation Report (week 10)
4.Sourcing components and building the lab-scale DC-DC converter prototype (week 4-15)	Sourced Component Inventory List (week 10)
	PCB Layout Files and Circuit Schematics (week 11)
	Fully Assembled Small-Scale DC-DC Converter Hardware (week 15)

GANTT CHART

EV Integrated PV System

Task Description	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Literature Review on EV PV Systems															
Designing Irradiance Meter & Testing Box															
Simulation & MPPT Algorithm															
Component Selection															
PCB Design & Hardware Assembly															
Final Testing & Bench Validation															

PV INTREGATED EV SYSTEM

System Integration and Range Anxiety

System Integration

The system consists of five main parts:

- Solar PV panel
- MPPT charge controller
- Converter
- Supercapacitor buffer
- Battery

The PV panel produces DC electricity from sunlight. As the panel output changes according to sunlight and temperature, we use a MPPT controller to keep the panel working at its best point to get the most power efficiently.

A DC-DC converter is used to raise the panel voltage.

Due rapid changes in sunlight, sudden current spikes caused. These spikes will affect the usage of battery which will wear it faster. Therefore a supercapacitor is placed between the MPPT controller and the battery to handle this kind of situation. The supercapacitor buffer absorbs these spikes before they reach the battery. These sudden changes cause current spikes that stress the battery and wear it out faster.

Adding a supercapacitor buffer can extend the battery life by 20 to 30 percent compared to connecting the solar panel directly to the battery.

For the prototype, a hybrid supercapacitor module is recommended because it can handle both energy storage and fast current response.



Battery Type Selection

According to the EV battery options across different vehicle types, the 48V LiFePO₄ pack, with 15 cells in series:

$$3.2\text{V} \times 15 = 48\text{V}, \quad 6\text{Ah per cell was selected for the prototype.}$$

This voltage works directly with standard MPPT solar charge controllers without any extra conversion. The **Battery Management System (BMS)** protects each cell by monitoring voltage, current, and temperature, and performs cell balancing to keep all cells at the same charge level.

Vehicle Type	Voltage	Capacity	Chemistry	Example
E-bicycle	24-48V	10-20Ah	Li-ion / LiFePO ₄	Hero Lectro
E-scooter	48-72V	24-60Ah	Li-ion / NMC	Ola S1, Ather 450X
E-motorcycle	72–120 V	30–80 Ah	NMC/LiFePO ₄	Ultraviolette F77
E-auto (3-wheel)	60-72V	60-120Ah	LiFePO ₄	Piaggio Ape E-City
Entry-level EV car	200-300V	20-40kWh	NMC / LFP	Tata Tiago EV
Mid-range EV car	350-400V	40-82kWh	NMC / LFP	Nexon EV, Ioniq 5

Table 01 : EV battery voltage reference

LiFePO₄ (Lithium Iron Phosphate) was selected because it is safer and lasts longer than other lithium battery types. It does not overheat easily which is better for our project. Other key features of this battery type are;

- Cycle life: 2000 to 4000 or more charge and discharge cycles
- Thermal safety: No thermal runaway risk. Works safely between -20 ° C and +60 ° C.
- Stable voltage: The cell voltage stays flat around 3.2V during most of the discharge, giving constant power output.

- BMS protection: The BMS checks each cell for over-voltage, under-voltage, over-current, and high temperature to stop damage before it happens

Battery health depends on how fast it charge and discharge, on its temperature, and how many cycles it has workout. According to studies it a good to maintain of discharge below 80% for LiFePO4 battery in solar systems to get the most cycles out of it. The converter also helps to improve battery health by keeping bus voltage stable even when solar input changes. This reduces voltage stress on the battery. The BMS also does cell balancing to make sure no single cell gets overcharged or over-discharged.

Reduction of Range Anxiety

Range Anxiety means the instance where EV runout of battery charge before coming to its destination. Now a days it is the most common problem to people for having doubt of buying electric vehicles. When using solar charging, this problem getting worse because we cannot get sunlight always throughout the day. So the amount of energy available to charge the battery changes every time. So we need to reduce this uncertainty much as possible to improve this technology.

Below are some design choices that directly reduce range anxiety in solar-charged EV systems:

- Right battery size: The prototype targets a 288Wh pack (48V, 6Ah) which suitable for light EVs.
- MPPT always maximizing energy: The MPPT controller makes sure the panel is always producing as much power as possible, even on cloudy days. This means the battery charges faster and more reliably.
- Using a Supercapacitor buffer for smooth charging: This reduce the wear out of battery by absorbing sudden spikes.
- Option for grid backup: Recommends designing the 48V bus so that it can accept both solar and grid charging. In this way, if there is not enough sun, the user can still charge from the grid and avoid getting stranded.

PV Panel Selection and MPPT Design

The system design deals with roof top installation area, irradiance measurement and performance under the changing environmental conditions such as shadows, clouds and different sunlight levels.

To obtain optimal energy output, the solar panel is selected based on its efficiency, size, and the required battery charging voltage. The panel is chosen by considering the trade-off between the available rooftop area and the desired energy output

Since PV output varies non linearly with irradiance and temperature, a MPPT technique is implemented. The MPPT controller continuously adjusts the operating point of the PV system. And ensuing that it operates at its maximum power point, regardless of environmental variations. This improves the overall efficiency and ensures stable energy delivery to the battery charging system.

PV Panel Selection

A flexible Mono-crystalline Silicon (Mono-Si) solar panel is selected for this considering Higher efficiency, Lightweight structure, Good flexibility for mounting and Lower degradation rate.

Estimated Rooftop Output Power

Parameter	Value
Solar Irradiance(per m ²)	1 kW/m ²
Panel Efficiency	20%
Estimated Area	1.5 m ²

Table 02

The solar power is given by,

$$P=A \times G \times \eta$$

Where:

P = Output power

A = Panel area

G= Solar irradiance

η = Panel efficiency

$$\begin{aligned}
 P &= A \times G \times \eta \\
 P &= 1.5 \times 1000 \times 0.20 \\
 &= 300 \text{ W}
 \end{aligned}$$

1.5 m² rooftop panel, can generate 300 W under standard test conditions.

Maximum Power Point Tracking(MPPT)

The output voltage and current of solar panels continuously change due to the environmental conditions. Because of this, direct connection between the PV panel and battery gives poor efficiency and energy loss.

MPPT controller is used to continuously track the maximum power point of the solar panel and provide maximum available energy to the boost converter.

MPPT ensures that the PV system always operates at its maximum power point by continuously adjusting the duty cycle of the boost converter. This improves energy efficiency under changing with varying environmental conditions.

Daily Energy Generation Calculation

PV system power = 300 W
Average peak sun hours in Sri Lanka $\approx$ 5 hours/day

$$\begin{aligned}
 E &= P \times t \\
 &= 300 \times 5 \\
 &= 1.5 \text{ kWh}
 \end{aligned}$$

So, Energy = 1.5 kWh/day

Added EV driving range

If EV consumes 80 Wh/km (typical threewheeler)

$$\begin{aligned}
 \text{Distance} &= \frac{1500 \text{ Wh}}{80 \text{ Wh/km}} \\
 &= 18.75 \text{ km}
 \end{aligned}$$

This represents additional daily range of 18.75 km provided by the solar charging system under average sunlight conditions.

Converter from PV to EV battery

The main goal is to interface a 300W peak flexible Mono-Si photovoltaic (PV) array to a 48V LiFePO4 battery bus. PV array produces 18V to 36 V, and the battery bus needs a stable 48 V, so it can't be connected directly to energy storage systems. Critical challenge here is the fluctuations of voltages in the PV system due to obstacles such as trees, clouds, buildings, and weather changes. A robust power conversion stage capable of stepping up the fluctuating input voltages and providing a stable 48 V is necessary.

Topology selection

A non-isolated boost DC-DC converter is selected for stepping up the voltage from 18 V to 48 V. Buck-boost or SEPIC topologies also provide the desired results, but they need a higher component count, like additional inductors and capacitors. They introduce high switching and conduction losses. The standard Boost converter provides the most direct step-up conversion with the lowest component count. Thereby minimising thermal losses and maximising energy transfer from the solar panels to the battery.

Parametric Calculations

1. The maximum duty cycle (D_{max})

It occurs under worst-case shading conditions, when the solar panel voltage drops to its minimum value,

$$V_{in(min)} = 18 \text{ V}$$

Using the boost converter duty cycle relation:

$$D_{max} = 1 - \frac{V_{in(min)}}{V_{out}}$$

$$D_{max} = 1 - \frac{18}{48}$$

$$D_{max} = 0.625$$

2. Inductor

The maximum input current ($I_{in(max)}$) must be determined to properly size the inductor.

Assuming ideal efficiency under worst-case operating conditions, the maximum output power of 300 W. The minimum input voltage of 18 V produces a maximum input current of:

$$I_{in(max)} = \frac{300}{18} \approx 16.67 \text{ A}$$

To maintain continuous conduction mode (CCM) operation and reduce excessive magnetic core losses, the design limits the inductor ripple current (ΔI_L) to 30% of the maximum input current:

$$\Delta I_L = 0.3 \times 16.67 \approx 5.0 \text{ A}$$

A switching frequency of 50 kHz is selected as a compromise between converter efficiency, switching losses, and passive component size. Higher switching frequencies reduce the required inductance and capacitance values. It results in a more compact design, but also increases switching losses in the MOSFET and diode.

The minimum required inductance is then calculated using:

$$L_{min} = \frac{V_{in(min)} \cdot D_{max}}{\Delta I_L \cdot f_{sw}}$$

Substituting the design values:

$$\begin{aligned} L_{min} &= \frac{18 \times 0.625}{5.0 \times 50000} \\ L_{min} &\approx 45 \mu\text{H} \end{aligned}$$

3. Capacitor

Finally, protecting the LiFePO₄ battery pack from high-frequency switching noise remains essential for maintaining battery health and system efficiency. The design limits the output voltage ripple (ΔV_{out}) to less than 1% of the 48 V output bus. Therefore, the maximum allowable ripple voltage is:

$$\Delta V_{out} = 0.01 \times 48 = 0.48 \text{ V}$$

The maximum output current at 300 W output power is:

$$I_{out(max)} = \frac{300}{48} \approx 6.25 \text{ A}$$

The minimum output capacitance required to satisfy the ripple specification is calculated as:

$$C_{out} = \frac{I_{out(max)} \cdot D_{max}}{f_{sw} \cdot \Delta V_{out}}$$

$$C_{out} = \frac{6.25 \times 0.625}{50000 \times 0.48}$$

$$C_{out} \approx 163 \mu\text{F}$$

By defining these theoretical parameters according to the 300 W maximum panel capacity, the design establishes a baseline inductance of $45 \mu\text{H}$ and a minimum output capacitance of $163 \mu\text{F}$. These component values provide the foundation required to construct and validate the open-loop power stage in the simulation environment.

The MOSFET and diode must withstand the maximum output voltage of 48 V and the peak inductor current. Therefore, devices with voltage ratings above 60 V and current ratings above 20 A are selected to ensure safe operation under transient conditions.

Under varying solar irradiance conditions, the boost converter maintains the DC bus voltage near 48 V by adjusting the duty cycle through the MPPT controller.

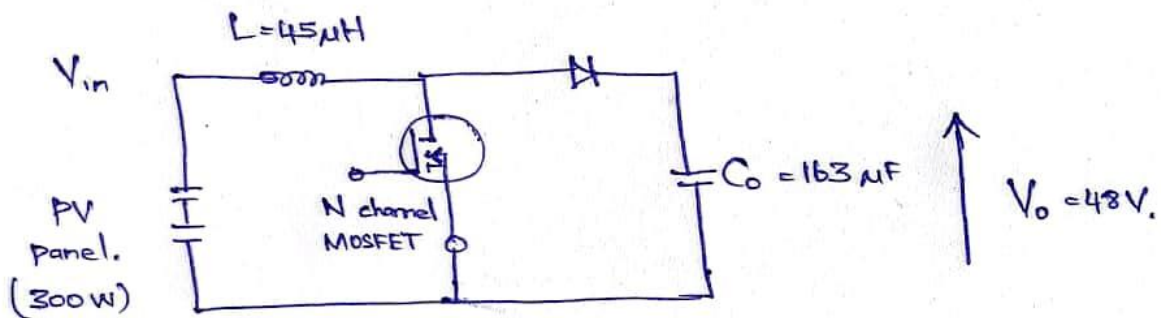


Figure 01: Boost converter topology with calculated values

REFERENCES

1. cntepower, “Battery Storage Price per kWh 2026: Full-Scenario Cost Analysis & Forecast - cntepower,” *cntepower*, Mar. 04, 2026. <https://en.cntepower.com/battery-storage-price-per-kwh-2026-full%e2%80%91scenario-cost-analysis-forecast/>
2. G. Eragamreddy and S. G. Naik, “Optimizing Electric Vehicle Range through Integrating Rooftop Solar on Vehicle,” *International Journal of Electrical and Electronics Research*, vol. 12, no. 1, pp. 66–72, Feb. 2024, doi: <https://doi.org/10.37391/ijeer.120111>.
3. Ahmad, Abdallah Odat, and A. Bashaireh, “A Novel PV Maximum Power Point Tracking Based on Solar Irradiance and Circuit Parameters Estimation,” *Sustainability*, vol. 14, no. 13, pp. 7699–7699, Jun. 2022, doi: <https://doi.org/10.3390/su14137699>.
4. A. Trivedi, V. Trivedi, and M. Gharib, “Choosing The Right Photovoltaic Panel for Electric Vehicles: An Integrated Decision Support Model,” *E3S Web of Conferences*, vol. 455, p. 02016, 2023, doi: <https://doi.org/10.1051/e3sconf/202345502016>.
5. A. R. Bhatti, Z. Salam, M. J. B. A. Aziz, K. P. Yee, and R. H. Ashique, “Electric vehicles charging using photovoltaic: Status and technological review,” *Renewable and Sustainable Energy Reviews*, vol. 54, pp. 34–47, Feb. 2016, doi: <https://doi.org/10.1016/j.rser.2015.09.091>.